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Assessing the spatial heterogeneous impacts of urban heat island effects on active travel by leveraging social media data

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ABSTRACT

This study investigates the impacts of urban heat island (UHI) effects on active travel by leveraging social media data. A multiscale geographically weighted regression (MGWR) model is utilized to investigate the spatial heterogeneity of integrated influences of UHI effects, built environment, and sociodemographic factors on travel frequency for both peri-summer and all-year trips. The investigation is showcased in Greater Melbourne, Australia, where Twitter posts related to active travel were collected and analyzed to identify active travelers' travel frequency in different suburbs. The results reveal that UHI effects had a significant negative impact on all suburbs, with greater intensity during peri-summer trips. Moreover, the results proved the spatial heterogeneity of the influence of UHI effects on active trips, with a more intensive influence in residential regions with high urban heat index values. Additionally, the density of tram stops, parkland areas, population density, and young adults had significant positive effects, while the unemployment rate and dwellings with one motor vehicle had negative impacts. This study contributes to the field of travel behavior analysis by completing location-contained social media data. Moreover, it identifies areas heavily impacted by UHI effects, enabling targeted measures such as expanding green spaces, using cooling materials, and enhancing energy practices to reduce UHI effects and promote a sustainable urban environment.

1. Introduction

Active travel, including walking and cycling, is a health-conscious and sustainable mode of transportation that offers significant benefits for both mental and physical well-being. Additionally, it helps reduce fossil fuel consumption and enhances the livability of neighborhoods. Despite the importance of active travel, the adoption of this green travel mode remains constrained, particularly in rural areas characterized by deficient pedestrian and cycling infrastructure, limited accessibility to destinations, and restricted land use diversity. In cities, active travel is more common due to well-connected roads, mixed land use, and easy access to various destinations. However, urbanization brings challenges that discourage active travel, such as higher risks of accidents with vehicles, more noise and air pollution, and environmental issues like the urban heat island effects. Among these challenges, the impact of the urban heat island effects on active travel has not been thoroughly studied.

Urban heat island (UHI) effects refer to the phenomenon of elevated temperature in urban areas compared to their surrounding regions, which mainly stem from alternations in built environment such as extensive utilization of low albedo building materials, dramatic reduction of vegetations, and heightened heat generation from power plants, vehicles, and air conditioners (Nuruzzaman, 2015).

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UHI effects lead to higher energy consumption for cooling buildings during summer, resulting in additional greenhouse gas emissions and air pollution from energy production (Santamouris, 2020). They also raise health risks, such as cardiovascular, respiratory, and cerebrovascular diseases (Piracha and Chaudhary, 2022). Moreover, UHI effects can influence people's travel behavior, particularly for active travelers who are directly exposed to heatwaves (Kamruzzaman et al., 2018). Existing research highlights the negative impact of high temperatures on active travel, showing that travel frequency, distance, and duration tend to decrease during hot seasons (Jia et al., 2022). However, no study has yet quantified the specific impact of UHI effects on the amount of walking and cycling at the city level. Filling this gap is crucial to indirectly comprehend the influences of microclimate changes introduced by urban development and human activity on active travel. This comprehension cannot be simply achieved by investigating the influences of high temperatures, as UHI effects are localized and persist longer, with nighttime temperature differences often being more pronounced (Filho et al., 2017), fundamentally altering active travel behavior in ways that regional climate conditions cannot fully explain.

Furthermore, due to the spatial disparities in urban form, green space distribution, and geographic conditions in different urban areas, the intensity of UHI effects can vary spatially, which in turn may introduce spatially varying impacts on active travel. Additionally, even in areas experiencing similar levels of UHI effects, differences in the built environment and sociodemographic characteristics can influence how UHI effects affect active travel. Despite these complexities, few studies have examined the spatial variability of UHI effects on active travel or accounted for the potential confounding effects of built environment and sociodemographic factors.

Understanding the influence of UHI effects requires a large, spatially disaggregated dataset, which is not often readily available. Traditionally, questionnaire surveys serve as the primary method for collecting travel behavior data (Zhou et al., 2022). Employing a local-scale travel survey emerges as an economical and efficient strategy for acquiring travelers' demographics and their travel behavior data, such as travel mode choices, travel frequency, and timing for travel behavior analysis. Nevertheless, it is noteworthy that data from such surveys typically stem from a limited participant pool within micro-scale spatial areas (Wu et al., 2021), where spatial variations of influential variables tend to be negligible. National travel surveys provide valuable datasets but are often costly and inefficient for collecting large-scale travel data. For example, Australia's national household travel survey, with a sample size of 5,100 households, is estimated to cost over 1 million AUD (Stopher et al., 2014). As a result, alternative data sources are needed to study the impact of UHI effects on active travel across large urban areas more efficiently.

Social media data (SMD) emerges as a promising alternative, partially alleviating the aforementioned challenges by offering substantial travel behavior data across expansive geographic domains at a comparatively lower cost (Hasnat and Hasan, 2018). Leveraging the prevalence of active social media users who routinely share their daily travels on platforms like Twitter and Facebook, researchers can effectively extract valuable travel-related insights. This includes information pertaining to travel purposes, destinations, modes of transportation (Rashidi et al., 2017), as well as sociodemographic details such as gender, age, education, and income levels (Sloan et al., 2015).

This study addresses the identified gaps by examining the impacts of UHI effects on active travel behavior using SMD to gather travel behavior insights. Specifically, it explores how the spatial heterogeneity of UHI effects impact the travel frequency of active travelers and considers the combined effects of built environment and sociodemographic factors. A case study is conducted in Melbourne, Australia, using Twitter data from 2018 to 2019. Tweets from pedestrians and cyclists are identified, and their activity locations are extracted from geotags and the content of tweets via natural language processing techniques. This data is used to compile a comprehensive record of active travel trips, including spatial and temporal details. To analyze the spatial variation in active travel trips, multiscale geographically weighted regression (MGWR) is applied, focusing on urban heat index values as the primary variable of interest, based on data from the Victoria Department of Transport and Planning (Victoria State Government 2023). The analysis also incorporates built environment and sociodemographic data as additional influencing factors.

The remaining sections of this paper are organized as follows: a literature review will be presented, followed by an introduction to the methodology to be utilized. Subsequently, a detailed description of a case study in Melbourne will be provided. Then, the main results and related analysis will be reported and discussed. Finally, the highlights, limitations, and future works of this study will be summarized and presented.

2. Literature review

Active travelers' travel behavior can be affected by diverse factors, which can be summarized as internal (e.g., self-selection effects) and external factors (e.g., built environment, weather). Internal factors refer to people's characteristics, preferences, and choices that affect their travel behavior, i.e., self-selection effects (Cao et al., 2009). For example, individuals with strong positive attitudes toward public transportation tend to choose neighborhoods with high transit accessibility (Hong et al., 2014). In such cases, their travel behavior is shaped more by their personal preferences than by the urban environment. External factors refer to environmental conditions that affect people's travel behavior, and these factors can be further divided into static and dynamic factors based on their temporal variability and stability (Lai and Kontokosta, 2018). Static factors are relatively constant and stable over time, such as built environment (diversity of land use, design of road network) and sociodemographic factors (Xu et al., 2025, Cheng et al., 2019, Wang et al., 2023). For example, people tend to use more active trips but travel shorter distances in urban areas with dense road networks and multiple types of land uses (Cheng et al., 2019). Dynamic factors are more variable and fluctuate over time, such as weather (Xu et al., 2021), and special events (e.g., sports events, concerts) (Eom and Nishihori, 2022, Li et al., 2017). For instance, large gatherings or events can lead to increased pedestrian volume and alter cycling routes (Eom and Nishihori, 2022).

Among these factors, weather plays a significant role in shaping active travel behavior. Under adverse weather conditions, people tend to reduce their travel time and distance, and adjust their travel modes accordingly (Xu et al., 2021, Zhu et al., 2023, Wu and Liao, 2020). As a key indicator of weather, temperature has been intensively investigated in studies on active travelers' behavior. High

temperatures make walking and cycling less comfortable, and can cause physical and physiological strain to pedestrians and cyclists, thus reducing their travel demands (Zhao et al., 2019, Zhao et al., 2018), and slowing their walking speed (Jia et al., 2022, Jia and Wang, 2021), especially during leisure trips (Wu and Liao, 2020). Analogous to but distinct from the effects of high temperature, the UHI effects refer to temperature deviations between urban and adjacent non-urban areas, which can impact people's mobility, such as affecting their travel distances, routes, and frequencies. Examining the influence of UHI effects on active travel behavior aids in comprehending heatwave effects on human health. However, research examining the relationship between UHI effects and active travel, particularly across areas with varying built environment and sociodemographic characteristics, remains limited.

There are two major challenges in identifying the potential spatially varying relationship between active travel and UHI effects. The first is selecting proper data sources that capture active travel behaviors across large-scale areas. The second is choosing a suitable model to analyze the spatial relationship between active travel and UHI effects. To address the first challenge, SMD provides a feasible solution. In previous studies examining the impacts of UHI effects on travel behavior, travel surveys have been the primary method for collecting information on individuals' travel patterns (Jia and Wang, 2021). However, these surveys are typically conducted within pre-selected areas, which may limit their applicability for analyzing the spatially varying impacts of factors like UHI effects on travel behavior. In contrast, SMD usually contains activity location information at the city or state level and can be applied for travel behavior analysis across broader geographic areas. For example, (Hasnat and Hasan, 2018) used geotagged Twitter data to reveal differences in the spatial distribution of travel destinations between local residents and tourists in Central Florida, United States. Similarly, (Plunz et al., 2019) analyzed geotagged tweets to estimate variations in travelers' sentiments between green spaces and non-green spaces in New York City, United States.

Geographically weighted regression (GWR) model (Özbil Torun et al., 2020) and multiscale geographically weighted regression (MGWR) model (Jia et al., 2022) are the two major adopted regression models to explore the spatial heterogeneous impacts of influencing factors on travel behavior. The GWR model can provide localized parameter (coefficient) estimates, i.e., offering regression coefficients for each spatial location (Fotheringham et al., 2002). In estimating the influencing factors' impact on active travel behavior for one location, the GWR model takes into account of influencing factors' values at both the location and its adjacent locations. Different distance weighting schemes, such as Gaussian kernel (Suryowati et al., 2021), bi-square kernel (Comber et al., 2023), and exponential kernel (Bidanset et al., 2017), can be employed to weight adjacent locations (Oshan et al., 2019), with higher weights given to locations that are closer in proximity. The bandwidth or spatial processing scale, considered an essential parameter in the GWR model, determines the number of surroundings to be included in the weighting scheme. A low bandwidth value represents a "local" relationship, where only nearby locations significantly affect the dependent variable at the observation site. Conversely, a high bandwidth value indicates a "global" relationship, where coefficients are estimated using observations from a wider surrounding area (Fotheringham et al., 2022).

An inherent constraint of the GWR model is that it assumes all explanatory variables are processed at the same bandwidth, which does not always hold true in real-world scenarios (Li et al., 2020). For instance, UHI effects are caused by the combination of multiscale factors, e.g., local built environment and global climate conditions. Simply utilizing the same bandwidth value for all explanatory variables may lead to biased coefficient estimation, as each variable may have its own unique optimal bandwidth.

The MGWR model is developed from the GWR model, inheriting the advantage of estimating localized coefficients, while releasing the bandwidth constraint by enabling independent adjustment of the bandwidths for different influencing variables (Fotheringham et al., 2017). Therefore, the MGWR model can explore both the local and global relationships between influencing factors and travel behavior. Due to the advantages of the MGWR model, it has been employed to investigate the impact of different factors on active travel. For instance, (Jia et al., 2022) explore the impact of neighborhood built environment on the walking behavior of old adults with the MGWR model, and find that the number of retail establishments, pedestrian crossings, and restaurants, are the most influential factors for the walking trip frequency of the elderly. In addition, (Li et al., 2022) investigate the association between the urban built environment and the ridership of dockless bike-sharing with the MGWR model. They find that the proximity to the central business district, the concentration of points of interest related to Hotels-Residences, and the density of the road network emerge as the most paramount determinants influencing bike-sharing ridership.

3. Methods

This study aims to explore the spatial heterogeneous impacts of UHI effects on active travel frequency. To this end, we employ SMD to gather travel behavior data and utilize the MGWR model for spatial regression analysis. Details of the data collection procedures and the key formulas used in the MGWR model are described below.

3.1. NLP-based data collection techniques

The framework of SMD collection is shown in Fig. 1 (Li et al., 2024). The travel behavior information can be captured through SMD by retrieving the posts that contain topic-related keywords. However, noises may be contained in these posts, and the target SMD needs to be selected. Text classification, a classic application of natural language processing (NLP) aiming at automatically categorizing text documents, can be utilized to select the target SMD. Among the extensive deep learning algorithms, the Bidirectional Encoder Representation from Transformers (BERT) model has demonstrated superior performance in text classification tasks due to its ability to capture the meaning of each word through its bidirectional surrounding contexts and the utilization of attention mechanism to weight the significance of each word in a sentence (Devlin et al., 2018). Therefore, this study utilizes a BERT model-based classifier (BERT-CLA) to select the topic-related (active travel-related) SMD.

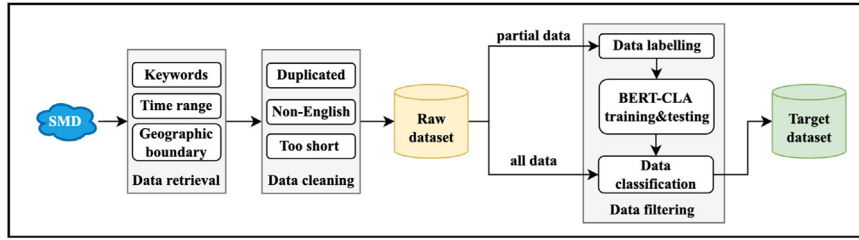


Fig. 1. Framework of social media data collection.

BERT is pre-trained on a large corpus using two key techniques (Devlin et al., 2018): a) Masked Language Model: Random words are masked, and the model predicts them to learn contextual relationships; b) Next Sentence Prediction: The model predicts whether one sentence logically follows another to understand sentence relationships. After pre-training, BERT is fine-tuned on labeled datasets for specific tasks, such as text classification, enabling it to achieve high accuracy and adaptability for task-specific requirements.

Activity location is a crucial type of travel behavior data, and the location entities embedded in the contents of SMD can be extracted through a named-entity-matching (NEM) technique. NEM is a method that matches word entities with location items of a location dictionary. The steps to implement NEM can be summarized as follows: 1) tokenization, this step is to separate a text sentence in word tokens; 2) location dictionary construction, this step is to build a dictionary that contains location items; 3) location entity matching, this step is to match the separated word tokens with items in the location dictionary. If the token matches a dictionary item, the token is labeled as a location entity. Activity location can also be extracted from geotags as it provides location coordinates. In cases of tweets with both content-based and geotag-based locations, only one type is chosen for travel behavior analysis. The selection prioritizes content-based over geotag-based, assuming higher accuracy of content-based locations since people typically mention the names of visited places in tweets but often geotag them elsewhere. Geotag-based location is an alternative for travel behavior analysis when the content-based location is absent.

3.2. MGWR model

In this study, the MGWR model is utilized to examine the influence of UHI effects on active travel behavior. The MGWR model can explore local relationships between the explanatory and response variables, assuming the relationships can be spatially varied (Fotheringham et al., 2017). In addition, the processing spatial scales (bandwidths) for relationships of different response and explanatory variables can be independently adjusted. The formula of the MGWR model is shown in Eq. (1).

$$y_i = \beta_0(u_i, v_i) + \sum_{j=1}^m \beta_{bwj}(u_i, v_i) x_{ij} + err_i \quad (1)$$

where y is the dependent variable, i.e., travel frequency of active travelers, (u, v) is the spatial coordination, β_0 is the intercept, $\beta_{bwj}(u_i, v_i)$ means the j th coefficient at location (u_i, v_i) , bw is the bandwidth, x_{ij} means the j th explanatory variable at location i , and err_i means error term at location i .

To maximize the performance of the MGWR model (Oshan et al., 2019), we use an adaptive bi-square kernel, as shown in Eq. (2) (Oshan et al., 2019), to weigh surrounding points for each observation point.

$$w_{ij} = \begin{cases} \left(1 - (d_{ij}/b)^2\right)^2 & \text{if } |d_{ij}| < b \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

where w_{ij} means the weight for the j th surrounding point of observation point i , d_{ij} means the distance between i and j , and b means the bandwidth.

Additionally, we utilize a golden section search optimization routine (Oshan et al., 2019) to determine the interval in the bandwidth optimization, and a corrected Akaike information criterion (AICc) to examine model fit performance, as shown in Eq. (3).

$$AIC_c = 2n \ln(RSS/n) + n \ln(2\pi) + n(n + tr(S))/(n + 2 - tr(S)) \quad (3)$$

where RSS means the residual sum of squares, n means the number of observations, and $tr(S)$ means the trace for the hat matrix (S).

The coefficients and bandwidths in the model are estimated via a back-fitting algorithm: optimize one parameter (one explanatory variable's coefficient and bandwidth) each time, assuming that all other parameters are known until all the parameters are optimized (Jia et al., 2022).

4. Case study

A case study was conducted to investigate the activities of active travelers in the 163 suburbs within the inner and middle Melbourne areas (Fig. 2). Melbourne is the capital of the state of Victoria, Australia, and it has 31 local government areas that can be grouped into three areas, each contributing to almost one-third of total jobs (Brain et al., 2019). The outer Melbourne area was excluded from the analysis due to the low number of active travel trips identified in SMD from that region.

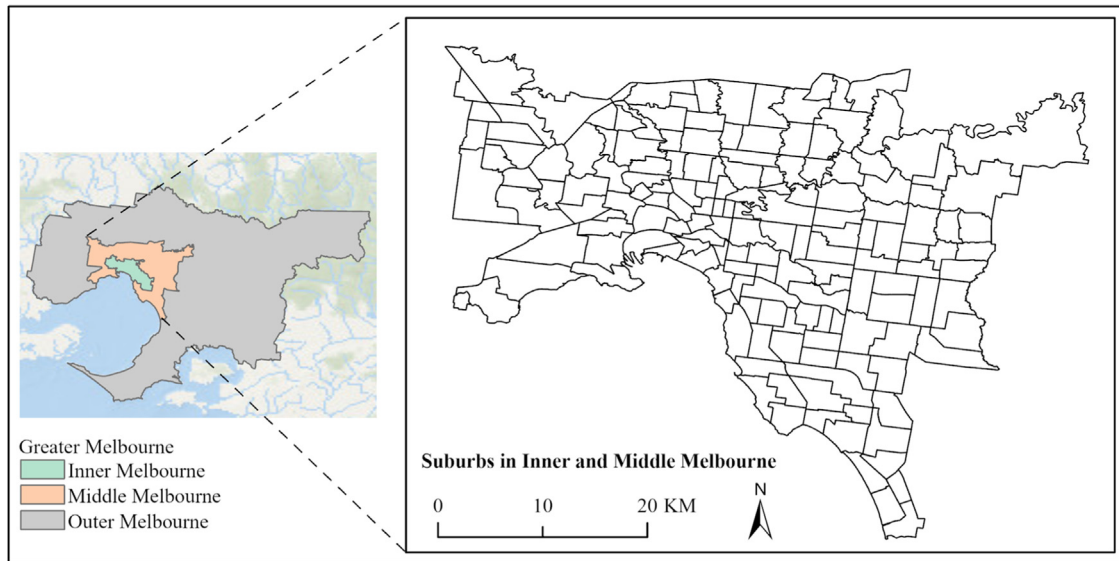


Fig. 2. Study area of suburbs in inner and middle Melbourne.

For training and validating the BERT-CLA model, a dataset of 1,000 manually labeled tweets was used, comprising 500 active travel-related tweets and 500 non-active travel-related tweets. Of this dataset, 80% was allocated for fine-tuning the BERT-CLA model, while the remaining 20% was reserved for validation. The fine-tuned BERT-CLA model demonstrated strong performance, achieving an accuracy of 0.85, a precision of 0.84, a recall of 0.87, and an F1-score of 0.85 in identifying active travel-related tweets. The performance of the NEM model was assessed by manually verifying the location information it extracted from 400 tweets. This evaluation included 200 tweets containing content-based location information and 200 without. The NEM model achieved an impressive accuracy of 95.74%, highlighting its effectiveness in accurately identifying content-based locations in tweets.

To explore the impacts of UHI effects on active travel, a dataset consisting of active travel data, urban heat data, built environment data, and sociodemographic data was constructed. Twitter posts (tweets) were selected as the data source for active travel data due to their good data availability compared with other social networks, such as Facebook and Instagram.

Steps to obtain target tweets and extract travel behavior information included: 1) collecting tweets containing active travel-related keywords (e.g., ‘walking’, ‘cycling’, ‘pedestrian’, ‘cyclist’) posted from 1 January 2018 to 31 December 2019 within inner and middle Melbourne area; 2) cleaning tweets to remove non-English and too short ones; 3) using the BERT-CLA to identify relevant tweets explicitly describing active travel activities, such as walking or cycling; 4) utilizing the NEM method to extract locations (content-based locations) from tweet texts; 5) integrating content-based locations with geotag locations (geotag-based locations) using a reasoning scheme to construct a location-contained tweet dataset; and 6) calculating the number of location-contained tweets in different suburbs to obtain the travel frequency of active travelers in these suburbs. Fig. 3 presents the distribution of active trips across all months of 2018–2019 (left panel) and specifically during the “peri-summer” period, spanning November to April (right panel). The involvement of trips during the peri-summer months aimed to examine whether the UHI effects were more pronounced during periods of elevated temperatures.

UHI data was obtained from the Victoria Department of Transport and Planning (Victoria State Government 2023). It provided Greater Melbourne’s urban heat index value (UHIV), i.e., the temperature deviation between a given urban area and its surrounding non-urban areas (native vegetated areas) using the urban temperature information collected by the United States Geological Survey. Due to varying urban built environment and geographic characteristics, UHIV varied significantly even within a small area (Fig. 4). Fig. 4a displays an example of the UHIV data given at the Mesh Block level. To prepare for regression analysis, the urban heat data were aggregated at the suburb level by weighing the area size of Mesh Blocks (Fig. 4b).

Built environment information was obtained from two sources: 1) land use information was provided by the Victoria Department of Transport and Planning, and the specific zoning information such as parkland zones and commercial zones were utilized; and 2) public transit stop information was obtained from DataVic (2023), and tram stop and bus stop locations were leveraged. Sociodemographic data, such as population number, age, gender, household income, employment status, education, and car ownership information, were obtained from the Australian Bureau of Statistics (2016) (Australian Bureau of Statistics 2023).

5. Results and discussion

5.1. Descriptive statistics

In this case study, data from 6 suburbs with very small populations (fewer than 100 residents) were excluded to minimize potential bias. As a result, only data from the remaining 157 suburbs were included in the modeling process. A total of 2,038 active travel trips, occurring between 2018 and 2019 within the 157 suburbs, were identified and used to construct the **AllTrip** dataset.

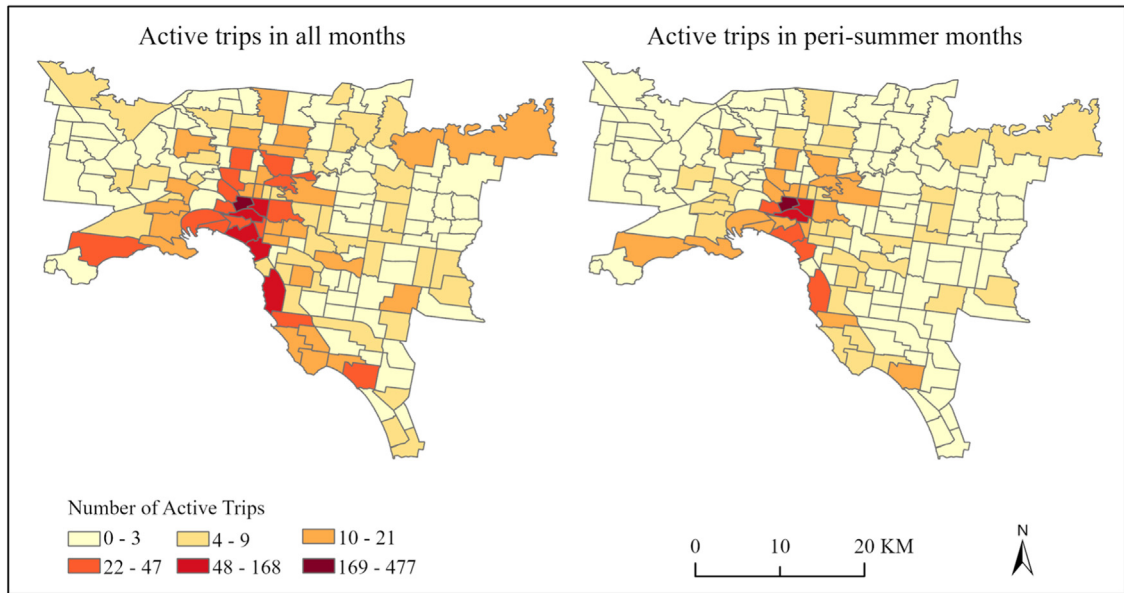


Fig. 3. Number of active trips in inner and middle Melbourne.

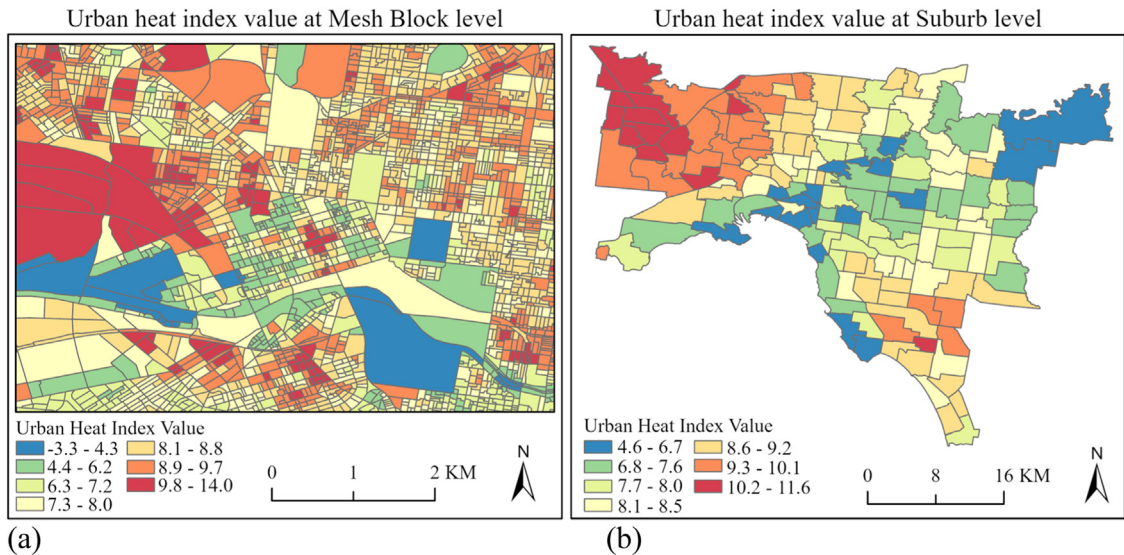


Fig. 4. Urban heat index values (2018).

The relatively small sample size of the AllTrip dataset is primarily due to the application of strict data filtering criteria. Tweets were required to fall within the defined spatial and temporal boundaries, be explicitly related to active travel, and most importantly, contain activity location information. While previous studies have successfully derived meaningful insights from datasets of similar size (Shirgaokar et al., 2021, Osorio-Arjona et al., 2021, Reynard and Shirgaokar, 2019), particular caution should be taken when conducting fine-grained analysis, as the limited sample may not adequately reflect detailed spatial and temporal patterns in active travel behavior.

For modeling preparation, active travel frequency, measured as the number of active trips in each suburb, was quantified by counting tweets containing location information within the respective suburb. For each trip, its location information was identified using either content-based location or geotag-based location. In the AllTrip dataset, 48.04% of trip locations were identified through content-based locations, while the remaining 51.96% were identified from geotag-based locations.

A seasonal subset comprising 1,082 trips occurring during the peri-summer period (November to April) was extracted to form the **SummerTrip** dataset. Within this subset, 43.72% of trip locations were identified using content-based locations, and 56.28% using geotag-based locations. Descriptive statistics for all variables across the 157 suburbs are presented in Table 1. Regression analyses

Table 1

Descriptive statistics of variables at the suburb level in the spatial modeling.

Category	Variable	Description	Mean	SD	Min	Max
Active travel trips [dependent variables]	AllTrip	Annual number of walk/cycle trips	12.99	42.33	0.00	477.00
	SummerTrip	Peri-summer number of walk/cycle trips	6.89	21.99	0.00	243.00
Urban heat island [key independent variable]	UHIV	Urban heat index value	8.38	1.25	4.67	11.61
Built environment [independent variables]	PopDen	Population density per square kilometer	3061.13	1.86	220.48	15764.08
	Residential	Ratio of residential areas	0.66	0.20	0.09	0.98
	Parkland	Ratio of parkland areas	0.16	0.12	0.00	0.66
	Commercial	Ratio of commercial areas	0.06	0.10	0.00	0.80
	TramStopDen	Density of tram stops per square kilometer	2.86	4.96	0.00	35.90
Sociodemographic factors [independent variables]	BusStopDen	Density of bus stops per square kilometer	10.22	4.19	0.95	21.20
	Female	Ratio of female	0.51	0.01	0.47	0.54
	Age0to14	Ratio of people aged 0-14 years	0.16	0.04	0.03	0.24
	Age15to34	Ratio of people aged 15-34 years	0.31	0.10	0.18	0.77
	Age35to64	Ratio of people aged 35-64 years	0.38	0.05	0.17	0.46
	Age65to84	Ratio of people aged 65-84 years	0.13	0.04	0.02	0.21
	Age85ov	Ratio of people aged 85 years and over	0.02	0.01	0.00	0.05
	HHIncome	Median household income per week	1670.19	345.39	561.00	2438.00
	UnempRate	Unemployment rate in people aged 15 years and over	0.07	0.03	0.04	0.19
	HighEdu	Ratio of people with bachelor's degrees and over	0.24	0.07	0.09	0.35
	0MotorVeh	Ratio of dwellings with no motor vehicle	0.11	0.10	0.01	0.75
	1MotorVeh	Ratio of dwellings with 1 motor vehicle	0.39	0.08	0.15	0.55
	2MotorVeh	Ratio of dwellings with 2 motor vehicles	0.35	0.09	0.03	0.48
	3MotorVeh	Ratio of dwellings with 3 motor vehicles	0.10	0.04	0.00	0.22
	4MotorVeh	Ratio of dwellings with 4 motor vehicles and over	0.05	0.03	0.00	0.19

were conducted on both the AllTrip and SummerTrip datasets to examine whether UHI effects have a stronger influence on active travel during the summer months.

A series of preliminary tests was carried out to assess the multicollinearity among various pairs of potential explanatory variables. For instance, the UHI effects stemming from urban development and human activities were scrutinized for their potential correlation with built environment (e.g., population density, ratio of parkland areas). Finally, 11 explanatory variables were chosen to investigate their impacts on active travel (Table 2), including urban heat island variable (urban heat index value), sociodemographic variables (population density, ratio of people aged 15-34 years, household income, unemployment rate, ratio of people with bachelor's degree and over, ratio of dwellings with 1 motor vehicle), and built environment variables (ratio of parkland areas, ratio of commercial areas, density of tram stops, density of bus stops). The Pearson correlation coefficients between each pair of these variables were below 0.80, and the variance inflation factor (VIF) values were all under 8, indicating there was no multicollinearity issue (Gao et al., 2022).

Table 2

OLS regression results predicting active travel trips.

Variable	Model: AllTrip		Model: SummerTrip	
	Coefficient	p-value	Coefficient	p-value
Intercept	29.031	0.467	16.246	0.434
UHIV	-6.474	0.010*	-3.669	0.005**
Parkland	0.890	0.000**	0.476	0.000**
Commercial	1.278	0.000**	0.670	0.000**
TramStopDen	3.846	0.000**	1.867	0.000**
BusStopDen	1.772	0.008**	0.836	0.015*
PopDen	6.997	0.003**	4.042	0.001**
Age15to34	-0.418	0.449	-0.248	0.390
HHIncome	0.029	0.052	0.016	0.047*
UnempRate	-2.409	0.150	-1.323	0.129
HighEdu	-2.224	0.007**	-1.199	0.006**
1MotorVeh	-0.834	0.024*	-0.379	0.048*

Adjusted R²: 0.62 for AllTrip, 0.62 for SummerTrip

* p-value<0.05,

** p-value<0.01

5.2. Modeling results

To ensure the bandwidths obtained from the MGWR model were comparable, both the explanatory and response variables were normalized according to Eq. (4). This normalization centered the variables at zero and aligned them within the same range of variation. Without this standardization, the amplitude of bandwidth may be associated with that of the explanatory variables, making the bandwidth comparison less objective. Then, for each trip dataset (AllTrip, SummerTrip), the global and local relationships between response variables (active trip frequency) and the selected explanatory variables were explored utilizing both the ordinary least squares (OLS) model and the MGWR model.

$$Var_{new} = (Var_{old} - \text{mean}(Var_{old})) / \text{std}(Var_{old}) \quad (4)$$

Examining global relationships provides a comprehensive understanding of the overarching impact of influencing factors on active travel. To this end, an OLS model was employed, and the corresponding regression results are presented in Table 2. Across both datasets, the majority of variables demonstrated significant impacts on active travel, with exceptions noted for household income, individuals aged 15-34 years, and the unemployment rate. To explore the spatial heterogeneity in the relationship between influencing factors and active travel, the MGWR model was utilized. Results indicated an enhanced Goodness of Fit for the MGWR model compared to the OLS model. Specifically, under the AllTrip dataset, the adjusted R-squared increased from 0.62 to 0.82, and under the SummerTrip dataset, it increased from 0.62 to 0.80. Given the MGWR model's superior performance, the subsequent sections focus on discussing its results, with no further elaboration on the OLS model.

The MGWR regression results of UHI effects on active travel frequency for two datasets are displayed in Fig. 5. UHI effects had significant negative impacts on both all-year trips (dataset All-year) and peri-summer trips (dataset SummerTrip). The intensity of negative effects was even stronger (lower negative coefficients) on peri-summer trips. This result indicated that individuals were more sensitive to UHI effects in summer seasons so they attempted to reduce exposure to a heat environment by decreasing uncomfortable active trips. In contrast, active trips were less affected in these non-summer seasons with moderate temperatures and lower UHI effects. The bandwidths of UHIV under the results of both datasets were approximated to the maximum bandwidth value (total number of suburbs, 157), indicating that nearly all neighboring suburbs were taken into the estimation of the coefficient of a local suburb. Such bandwidth revealed that the relationship between UHI and active travel was globally stationary, i.e., UHI effects had negative influences on active travel in all suburbs.

Our findings also revealed a significant spatial heterogeneity in the impacts of UHI effects on active travel, as illustrated in Fig. 5. Notably, the impact exhibited a partial alignment with the spatial distribution pattern of urban heat index values (UHIVs), as depicted in Fig. 4b. Specifically, stronger negative impacts of UHI effects were observed in the northwest regions, corresponding to the highest UHIVs in these areas. In contrast, middle regions, traversed by the Yarra River and characterized by lower UHIVs, exhibited a weaker impact on active travel. The observed coherence between the impact intensity of UHI effects and the temperature elevation in urban areas is logical, indicating a direct correlation between the degree of higher temperatures and the influence on people's active travel patterns. However, noteworthy discrepancies emerged between the impact of UHI effects on active travel and the spatial distribution pattern of UHIVs. In detail, the extent of UHI effects' impact diminished from the middle to the southeast regions, presenting an opposing trend to the variation in UHIVs. A comprehensive analysis of the Google Satellite Map revealed that the southeastern regions, characterized by significantly elevated UHIVs, were predominantly composed of industrial land uses, including facilities

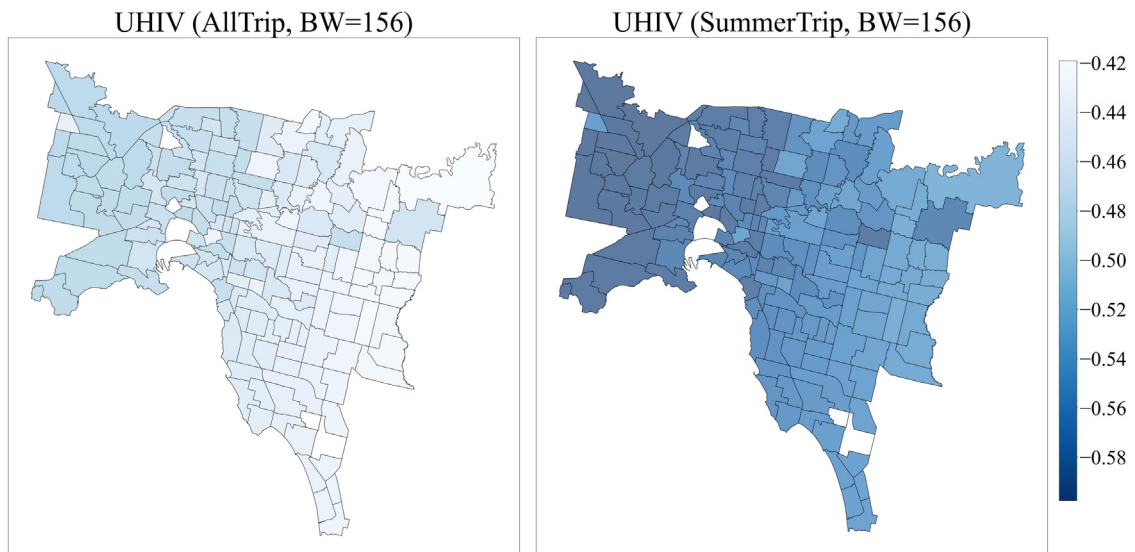


Fig. 5. MGWR regression results of UHI effects for two datasets (BW means bandwidth, white colored polygons represent suburbs with less than 100 people).

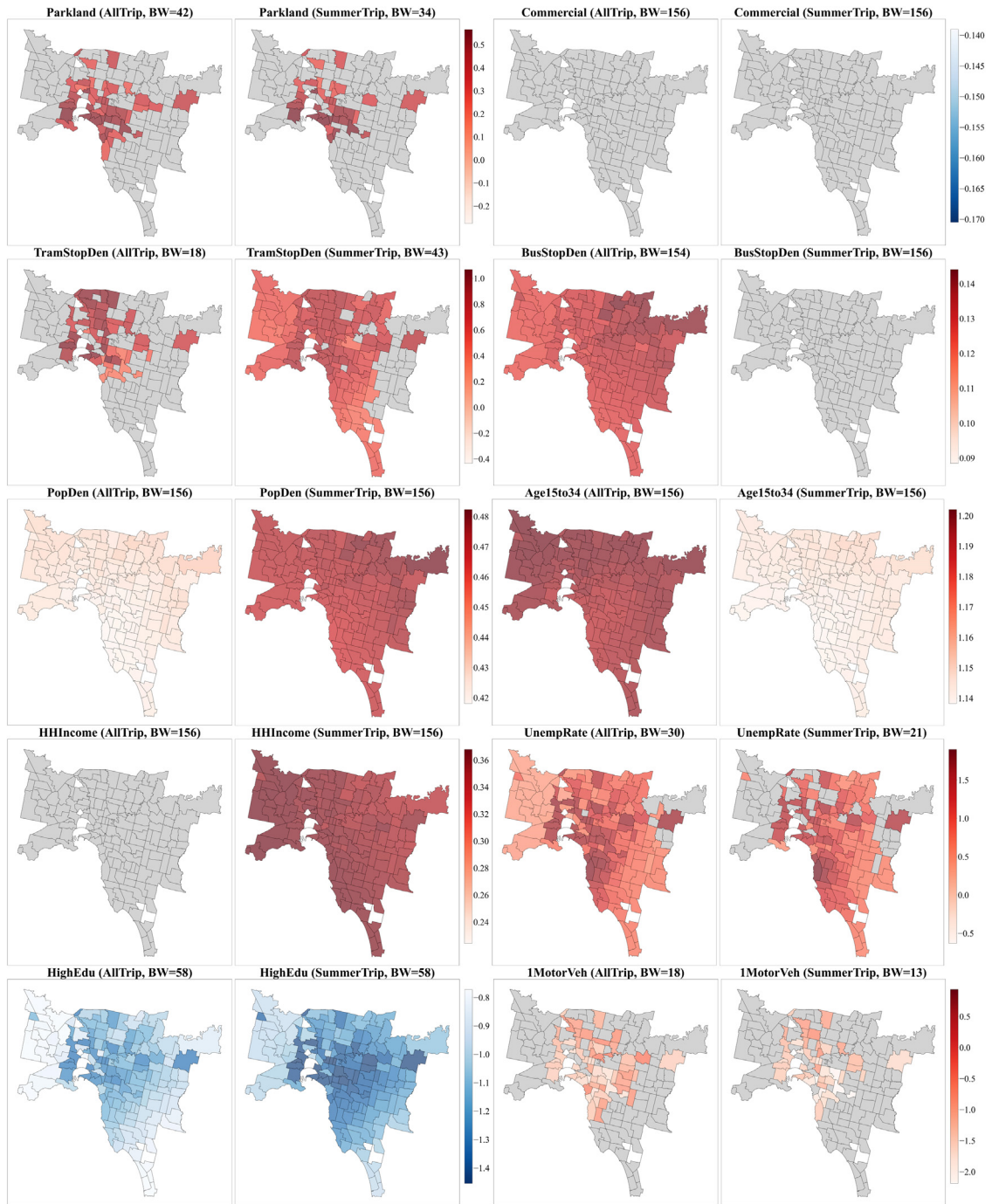


Fig. 6. MGWR regression modeling results of resting variables for two datasets (grey colors represent results that are insignificant at the level of 0.05).

such as airports, flying schools, manufacturing plants, and breweries. These areas typically see minimal walking or cycling activity, resulting in negligible UHI effects. In contrast, the northwestern regions were primarily residential, with a higher likelihood of active travel, making them more susceptible to UHI effects. In summary, the spatial heterogeneity in UHI effects on active travel was closely linked to both UHIV levels and the prevailing land use patterns.

The MGWR regression results of the remaining influential factors are shown in Fig. 6. The density of tram stops demonstrated significant positive impacts on both all-year and peri-summer trips. This result can be explained by regions with extensive tram stops providing pedestrians and cyclists with easier access to adjacent places such as libraries, schools, and shopping centers. As a

result, active trips increased from these tram stops to adjacent destinations and vice versa. Additionally, compared to all-year trips, the density of tram stops showed diminished (lower coefficients) impacts on peri-summer trips. This finding implied a weaker link between tram stop accessibility and active travel during summer, possibly attributed to people's tendency to reduce their exposure to heat waves. Regarding the suburbs where the density of tram stops showed stronger impacts (higher coefficients), it was found that these areas substantially overlapped with suburbs featuring a high concentration of tram stops. Furthermore, the bandwidth value under the SummerTrip model was significantly higher than that under the AllTrip model, reflecting seasonal changes in travel behavior. During summer, people often adjusted their routines to avoid heat exposure, which reduced the immediate impact of tram stop accessibility on active travel. For example, individuals tended to stay indoors, use alternative transportation, or travel to destinations that did not require walking or cycling. In areas with high tram stop density, these adjustments led to fewer localized trips tied to tram stops, weakening their local influence. To capture meaningful trends under these conditions, data from a broader area, including farther suburbs, became relevant. This wider spatial integration accommodated the more dispersed and sporadic travel patterns typical of summer but also included less relevant influences, as tram stops in distant suburbs were unlikely to significantly affect local active travel behavior.

The density of bus stops exhibited significant positive impacts on all-year trips but nonsignificant impacts on peri-summer trips. Bus stops and tram stops were perceived to fulfill comparable roles in facilitating active travel by enhancing accessibility, particularly for pedestrians, to surrounding locales. However, there were distinct differences in their spatial distributions. Tram stops were mainly concentrated in central suburbs, while the bus stops were more evenly distributed across all suburbs, resulting in a global stationary impact of density of the bus stops on active travel (large bandwidth and spatial distribution of coefficients were consistent). The absence of significant impacts of bus stop density on peri-summer trips could potentially be attributed to the strong competition posed by tram stops.

The household income showed nonsignificant impacts on all-year trips but significant positive impacts on peri-summer trips. Although certain studies suggested that households with higher income levels tended to possess more opportunities to acquire private cars, consequently leading to a reduction in the use of active travel options (Ek et al., 2021), our study supported an alternative scenario, i.e., high-income families may tend to use more active travel due to their conscientious efforts in maintaining physical and mental health (Khan et al., 2014).

The unemployment rate presented significant negative impacts on both all-year trips and peri-summer trips. This result can be explained as that unemployed people utilized fewer commuting active trips (Yu et al., 2023). Consequently, in centrally located suburbs with convenient access to workplaces, an elevated unemployment rate exacerbated the negative influences on active travel frequency, as indicated by the lower negative coefficients observed in these areas.

People with higher education levels (with bachelor's degrees and over) exhibited significant negative impacts on both all-year trips and peri-summer trips. This result contradicted certain studies suggesting that people with higher education levels tend to adopt non-motorized modes of transportation such as walking and cycling (Wu et al., 2021, Plaut, 2005). Our analysis of the raw data revealed a lack of substantial overlap between suburbs exhibiting a high number of active trips and those characterized by a higher prevalence of highly educated residents. For instance, people visit the suburb of the central business district (CBD) for diverse purposes such as working, shopping, and traveling, resulting in an extremely large number of active trips in both the CBD and its adjacent suburbs. However, people with higher education levels were not solely concentrated in these few suburbs; rather, they lived more evenly in both central and peripheral suburbs. As a result, the majority of suburbs with substantial numbers of well-educated people did not demonstrate markedly higher active travel frequency, contributing to the observed negative correlation between active travel and higher education level. This finding indicated that the places with extremely high numbers of active trips did not necessarily align with those housing a higher density of well-educated individuals, as people preferred to live in non-central areas for more affordable housing and a more livable environment.

Households with one motor vehicle showed significant negative effects on both all-year trips and peri-summer trips, and the negative effects were particularly conspicuous in central suburbs. This result can be attributed to that the competitiveness between active travel and private cars increased when people live or work closer to the city center (Tennøy et al., 2022). City centers usually have road networks and diverse land uses, facilitating the preference for active travel over shorter distances (Wu et al., 2021). However, research also indicated that family with private cars tends to use less active travel (Yang et al., 2022). As a result, the competition between the two modes (active travel versus private car) was more profound in central suburbs. Some other variables also showed significant effects on active travel. For instance, parkland areas, population density, and people aged from 15 to 34 showed significant positive effects.

6. Conclusion

This study explored the spatial heterogeneity of the impact of urban heat island effects on active travel behavior in Inner and Middle Melbourne. By utilizing geotag and content information from Twitter data, we obtained travel frequency data of pedestrians and cyclists in different suburbs. Subsequently, we employed a multiscale geographically weighted regression model to investigate the spatially varying impacts of UHI effects, built environment factors (e.g., land use, tram and bus stop density), and sociodemographic factors (e.g., age, education, income, car ownership, unemployment rate) on active travel. The results indicated significant negative impacts of UHI effects on active travel frequency, with a stronger effect observed during peri-summer trips. Significantly, the influence of UHI effects was observed to exhibit heterogeneity across various regions, displaying a more pronounced effect on residential zones with elevated urban heat index values. Furthermore, the density of public transit stops, particularly tram stops, positively influenced active travel, suggesting individuals' preference for utilizing public transit to connect to nearby amenities such as libraries, schools,

and parks. Additional significant factors included the percentage of parkland areas and population density (positive effects), the percentage of people aged 15-34 (positive effects), the unemployment rate (negative effects), and the percentage of dwellings with one motor vehicle (negative effects).

This study integrates content-based and geotag-based location information to analyze active travel behavior, resulting in a richer dataset and enhancing the robustness of the analysis. Furthermore, it innovatively examines the spatial heterogeneity of UHI effects on active travel frequency. The findings provide valuable insights into how urban development and human activities, such as the extensive use of low-albedo materials, vegetation loss, and heat generated by power plants, vehicles, and air conditioners, impact travel behavior. These insights can assist urban planners and transportation stakeholders in optimizing urban design, replacing building materials, and promoting energy usage habits that foster a more active travel-friendly environment. Additionally, as regions with heightened UHI impacts have been identified, targeted interventions can be implemented, such as increasing green spaces and implementing cooling pavements. Although conducted pre-pandemic, the study's findings remain highly relevant for post-pandemic policymaking aimed at enhancing urban livability and sustainability. Addressing UHI-related barriers is crucial for improving public health, mobility, and the promotion of active travel in the post-pandemic era.

Despite these contributions, this study has several limitations. First, the dataset is relatively small, consisting of 2,038 trips collected over a two-year period. This limited sample size reduces the ability to capture fine-grained spatial and temporal patterns in active travel, resulting in a more qualitative rather than quantitative analysis. Second, the use of Twitter data introduces inherent demographic bias. Social media platforms often overrepresent digitally active populations, typically younger, higher-income, and urban residents. This demographic imbalance limits the generalizability of the findings to the broader population. Third, some bandwidth values produced by the MGWR model were excessively large and unrealistic, deviating significantly from expected spatial influences for variables such as tram stop density during peri-summer trips. This undermines the practical interpretability of those results. Future research can enhance sample size by incorporating data from a broader range of social media platforms, such as Facebook and Instagram, and by extending the data collection period, such as including data from the years 2016–2017. To mitigate demographic bias, text mining techniques can be employed to infer and supplement key sociodemographic attributes of social media users, such as age, gender, and education level. Finally, applying constraints to bandwidth selection in the MGWR model can enhance the clarity and contextual relevance of the estimated effects.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Teng Li: Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Conceptualization. **Zhuo Chen:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization. **Shuli Luo:** Writing – review & editing, Methodology. **Alexa Delbos:** Writing – review & editing, Investigation, Conceptualization.

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